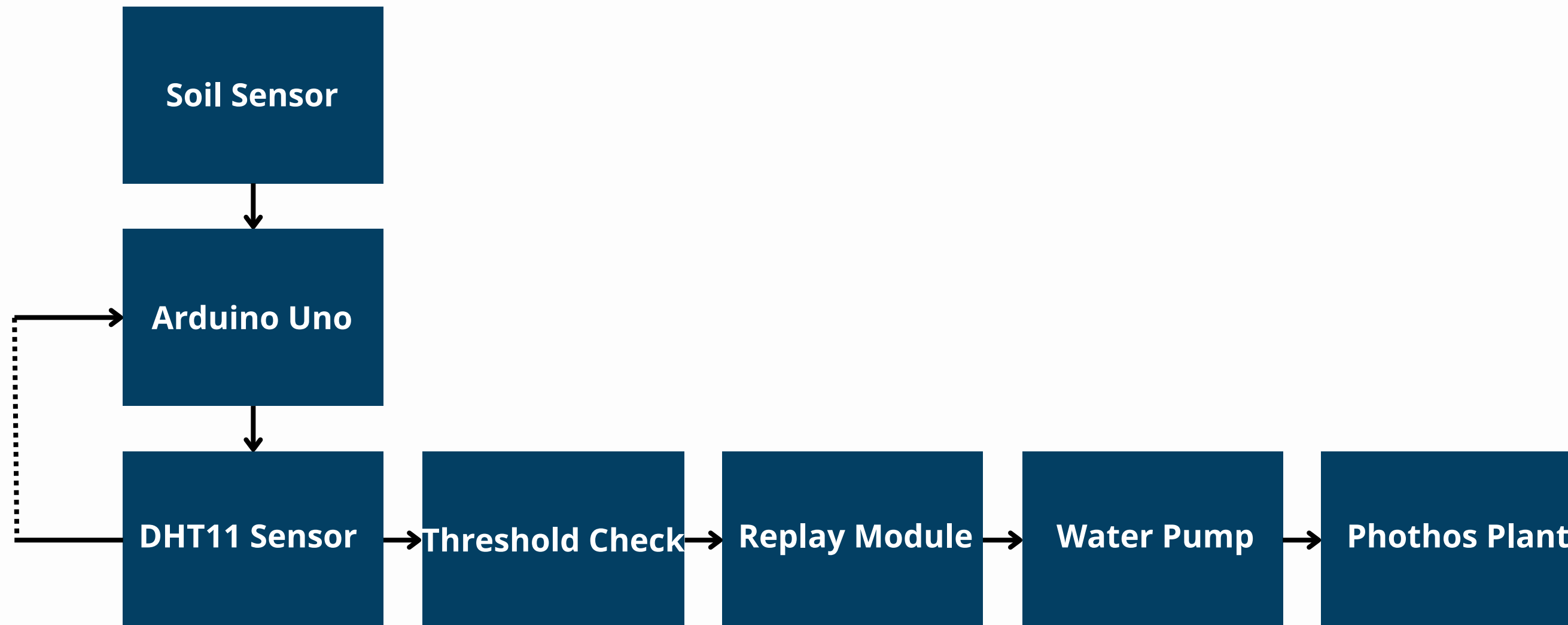


SYSTEM ARCHITECTURE

WORKING PROCESS

1. SOIL MOISTURE SENSOR AND DHT11 COLLECT DATA EVERY 5 MINUTES.
2. ARDUINO UNO PROCESSES SENSOR READINGS.
3. SOIL MOISTURE FALLS BELOW THRESHOLD (35–40%).
4. RELAY MODULE IS ACTIVATED.
5. MINI WATER PUMP SUPPLIES WATER.
6. PUMP STOPS AUTOMATICALLY WHEN OPTIMAL MOISTURE IS REACHED.

SYSTEM ARCHITECTURE



HARDWARE IMPLEMENTATION

Component	Function
Arduino Uno R3	Main controller
Soil Moisture Sensor	Measures soil moisture
DHT11 Sensor	Measures temperature & humidity
Relay Module	Controls pump switching
Mini Water Pump	Waters the plant
Breadboard & Jumper Wires	Circuit connections
Pipe	Pipe Water delivery

HARDWARE IMPLEMENTATION

